Dictionary

Add new key-value

In [1]:
```
alien_0={"color":"green","points":5}

print(alien_0['color'])
print(alien_0['points'])
```

```
green
5
```

In [2]:
```
print("You just earned "+ str(alien_0['points'])+" points!")
```

```
You just earned 5 points!
```

In [3]:
```
alien_0['points']=10
print(alien_0)

print("You just earned "+ str(alien_0['points'])+" points!")
```

```
{'color': 'green', 'points': 10}
You just earned 10 points!
```

增加新key 和 value必须要加符号''(数字除外)

In [4]:
```
alien_0['x_position']=0
alien_0['y_position']=25
print(alien_0)
```

```
{'color': 'green', 'points': 10, 'x_position': 0, 'y_position': 25}
```

Start with empty dict

In [5]:
```
alien={}
alien['color']='green'
alien['points']=5
print(alien)
```

```
{'color': 'green', 'points': 5}
```

Modifying value

In [6]:
```
alien['color']='blue'
print(alien)
```

```
{'color': 'blue', 'points': 5}
```

In [7]:
```
alien_0={"color":"green","points":5}
print("This alien's color is:" + alien_0['color'])

alien_0['color']='blue'
print("This alien's color is:" + alien_0['color'])
```

```
This alien's color is:green
This alien's color is:blue
```

In [8]:
```
alien_0={"x_position":0,"y_position":10,"speed":"medium"}
print("Original x_position: "+str(alien_0['x_position']))

if alien_0["speed"]=='slow':
    x_increment = 1
elif alien_0["speed"]=='medium':
    x_increment = 2
else:
    x_increment = 3

alien_0["x_position"]=alien_0["x_position"]+x_increment
print("New x_position: " + str(alien_0["x_position"]))
```

```
Original x_position: 0
New x_position: 2
```

In [9]:
```
# 薄弱题
alien_0={"x_position":0,"y_position":10,"speed":"medium"}
alien_0["speed"]="fast"
print("Original x_position: "+str(alien_0['x_position']))

if alien_0["speed"]=='slow':
    x_increment = 1
elif alien_0["speed"]=='medium':
    x_increment = 2
else:
    x_increment = 3

alien_0["x_position"]=alien_0["x_position"]+x_increment
print("New x_position: " + str(alien_0["x_position"]))
```

```
Original x_position: 0
New x_position: 3
```

Removing Key-Value Pairs--del

In [14]:
```
alien_0 = {'color': 'green', 'points': 5}
print(alien_0)
```

```
{'color': 'green', 'points': 5}
```

In [15]:
```
del alien_0["color"]
print(alien_0)
```

```
{'points': 5}
```

In [16]:
```
del alien_0["points"]
print(alien_0)
```

```
{}
```

Excercise

In [17]:
```
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}
print("Sarah's favorate language is " + favorite_languages['sarah'].title()+".")
```

```
Sarah's favorate language is C.
```

In [19]:
```
for k,v in favorite_languages.items():
    print(k.title()+"'s favorate language is "+ v.title() +"!")
```

```
Jen's favorate language is Python!
Sarah's favorate language is C!
Adward's favorate language is Rstudio!
Phli's favorate language is Python!
```

In [20]:
```
# 6-1
person={
    "first_name":"Mica",
    "last_name":"Liang",
    "age":25,
    "city":"Guangxi"
}

for k,v in person.items():
    print("\nKey:"+k)
    print("Value:"+str(v))
```

```
Key:first_name
Value:Mica

Key:last_name
Value:Liang

Key:age
Value:25

Key:city
Value:Guangxi
```

In [21]:
```
# 6-2
favorate_num={
    "mica":9,
    "donna":5,
    "wang":6,
    "sara":7
}

for k,v in favorate_num.items():
    print(k.title()+"'s favorate number is "+ str(v) +".")
```

```
Mica's favorate number is 9.
Donna's favorate number is 5.
Wang's favorate number is 6.
Sara's favorate number is 7.
```

In [22]:
```
# 6-3
glossary={
    "list":"Can store different types of data",
    "dictionary":"Stores information using key-value pairs",
    "while":"Used to create loops",
    "if":"Used to make decisions",
    "flag":"Used to control loops"
}

for k,v in glossary.items():
    print(k.title()+": "+ v)
```

```
List: Can store different types of data
Dictionary: Stores information using key-value pairs
While: Used to create loops
If: Used to make decisions
Flag: Used to control loops
```

looping dictionary

Looping all Key-Value Pairs

In [25]:
```
user={
    "username":"efermi",
    "first":"enrico",
    "last":"fermi"
}

for k,v in user.items():
    print("\nKey: "+k)
    print("Value: "+v)
```

```
Key: username
Value: efermi

Key: first
Value: enrico

Key: last
Value: fermi
```

Looping all Key

In [90]:
```
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}

for k in favorite_languages.keys():
    print(k.title())
```

```
Jen
Sarah
Adward
Phli
```

In [28]:
```
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}

friends=['phli','sarah']
for k in friends:
    print(" Hi " +k.title()+", I see your favorate language is "+favorite_langua
```

```
Hi Phli, I see your favorate language is Python!
Hi Sarah, I see your favorate language is C!
```

In [135…
```
# 薄弱题
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}

friends=['phli','sarah']
for key in favorite_languages.keys():
    print(key.title())

    if key in friends:
        print(" Hi " +key.title()+", I see your favorate language is "+favorite_

if "erin" not in favorite_languages.keys():
    print("\nErin,please take our poll!")
```

```
Jen
Sarah
 Hi Sarah, I see your favorate language is C!
Adward
Phli
 Hi Phli, I see your favorate language is Python!

Erin,please take our poll!
```

In [29]:
```
foods={
    "jenny":"fish",
    "john":"beef",
    "sunny":"lamb",
    "jack":"egg"
}
friends=["john","jack"]

for key in foods.keys():
    print(key.title())
    if key in friends:
        print(" Hi "+key.title()+"'s favorite food is "+foods[key].title())
```

```
Jenny
John
 Hi John's favorite food is Beef
Sunny
Jack
 Hi Jack's favorite food is Egg
```

sorted()--Looping Keys in Order

In [145…

```python
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}

for name in sorted(favorite_languages.keys()):
    print(name.title() + ", thank you for taking the poll.")
```

```
Adward, thank you for taking the poll.
Jen, thank you for taking the poll.
Phli, thank you for taking the poll.
Sarah, thank you for taking the poll.
```

In [31]:

```python
foods={
    "candy":"fish",
    "anna":"beef",
    "bruce":"lamb",
    "delly":"egg"
}

for key in sorted(foods.keys()):
    print("Hi "+key.title()+", i see your favorite food is "+foods[key]+"!")
```

```
Hi Anna, i see your favorite food is beef!
Hi Bruce, i see your favorite food is lamb!
Hi Candy, i see your favorite food is fish!
Hi Delly, i see your favorite food is egg!
```

sorted()--Looping value in Order

In [33]:

```python
# 重复也会被全部输出
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}

for v in sorted(favorite_languages.values()):
    print(v.title())
```

```
C
Python
Python
Rstudio
```

dict + set() 避免重复输出

In [153…

```
# 薄弱题
# 用set处理，避免重复输出
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}

for v in set(favorite_languages.values()):
    print(v.title())
```

```
Python
Rstudio
C
```

Excercise

In [34]:

```
# 6-4
glossary={
    "list":"Can store different types of data",
    "dictionary":"Stores information using key-value pairs",
    "while":"Used to create loops",
    "if":"Used to make decisions",
    "flag":"Used to control loops"
}

for k,v in glossary.items():
    print(k.title()+": "+ v)
```

```
List: Can store different types of data
Dictionary: Stores information using key-value pairs
While: Used to create loops
If: Used to make decisions
Flag: Used to control loops
```

In [35]:

```
# 6-5
rivers={
    'nile': 'egypt',
    'amazon': 'brazil',
    'yangtze': 'china'
}

for k,v in rivers.items():
        print("The "+k.title()+" runs through "+rivers[k]+"!")
```

```
The Nile runs through egypt!
The Amazon runs through brazil!
The Yangtze runs through china!
```

In [172…

```
rivers={
    'nile': 'egypt',
    'amazon': 'brazil',
    'yangtze': 'china'
```

```
}

for k,v in rivers.items():
    print("\nRiver_name:"+k)
    print("Country_name:"+v)
```

```
River_name:nile
Country_name:egypt

River_name:amazon
Country_name:brazil

River_name:yangtze
Country_name:china
```

In [42]:

```
# 6-6
favorite_languages={
    'jen':'python',
    'sarah':'c',
    'adward':'rstudio',
    'phli':'python'
}
poll_names=["sarah","phli"]

for key in favorite_languages.keys():
    print(key.title())

    if key in poll_names:
        print(" -Hi "+key.title()+", thank you for your responding.")
    if key not in poll_names:
        print(" -Hi "+key.title()+", please polling.")
```

```
Jen
 -Hi Jen, please polling.
Sarah
 -Hi Sarah, thank you for your responding.
Adward
 -Hi Adward, please polling.
Phli
 -Hi Phli, thank you for your responding.
```

list and dictionary nesting(嵌套)

dict in list

In [3]:

```
# dictionary in list
alien_0={"color":"green","points":5}
alien_1={"color":"yellow","points":10}
alien_2={"color":"red","points":15}

aliens=[alien_0,alien_1,alien_2]

for alien in aliens:
    print(alien)
```

```
{'color': 'green', 'points': 5}
{'color': 'yellow', 'points': 10}
{'color': 'red', 'points': 15}
```

dict + range() + list--批量创建字典

In [6]:
```python
aliens=[]

for alien_num in range(30):
    new_alien={"color":"green","points":5,"speed":"slow"}
    aliens.append(new_alien)

for alien in aliens[:5]:
    print(alien)
print("...")

print("Total number of the aliens are: "+str(len(aliens)))
```

```
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
...
Total number of the aliens are: 30
```

dict + range() + list + for--修改字典

In [45]:
```python
# Make an empty list for storing aliens.
aliens=[]

# Make 30 green aliens.
for alien in range(30):
    new_alien={'color': 'green', 'points': 5, 'speed': 'slow'}
    aliens.append(new_alien)

# Show the first 5 aliens:
for alien in aliens[0:5]:
    print(alien)
print("...")

# to change the first 3 aliens
for alien in aliens[0:3]:
    if alien['color']=='green':
        alien['color']='blue'
        alien['speed']='medium'
        alien['points']=10

# output the update aliens
print("Update aliens:")
for alien in aliens[0:5]:
    print(alien)
```

```
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
...
Update aliens:
{'color': 'blue', 'points': 10, 'speed': 'medium'}
{'color': 'blue', 'points': 10, 'speed': 'medium'}
{'color': 'blue', 'points': 10, 'speed': 'medium'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
{'color': 'green', 'points': 5, 'speed': 'slow'}
```

In [31]:

```python
# 薄弱题
# crate a empty list for storing
students=[]

# create 20 student info
for student in range(20):
    student_info={'name':'student','grade':80,'class':1}
    students.append(student_info)

# output first 5 students info
for student in students[0:5]:
    print(student)
print("...")

for student in students[0:3]:
    if student['grade']==80:
        student['grade']=95
        student['class']=2

print("student info update:")
for student in students[0:5]:
    print(student)
print("...")
```

```
{'name': 'student', 'grade': 80, 'class': 1}
{'name': 'student', 'grade': 80, 'class': 1}
{'name': 'student', 'grade': 80, 'class': 1}
{'name': 'student', 'grade': 80, 'class': 1}
{'name': 'student', 'grade': 80, 'class': 1}
...
student info update:
{'name': 'student', 'grade': 95, 'class': 2}
{'name': 'student', 'grade': 95, 'class': 2}
{'name': 'student', 'grade': 95, 'class': 2}
{'name': 'student', 'grade': 80, 'class': 1}
{'name': 'student', 'grade': 80, 'class': 1}
...
```

In [47]:

```python
# create an empty list to store
items=[]

# create 25 items
for item in range(25):
    item_info={'type': 'potion', 'power': 10, 'price': 50}
    items.append(item_info)

# output first 5 items
```

```
for item in items[0:5]:
    print(item)
print("...")

# update information
for item in items[0:2]:
    if item['type']=='potion':
        item['type']='sword'
        item['power']=50
        item['price']=200

# output the update itemds
for item in items[0:5]:
    print(item)
print("...")
```

```
{'type': 'potion', 'power': 10, 'price': 50}
{'type': 'potion', 'power': 10, 'price': 50}
{'type': 'potion', 'power': 10, 'price': 50}
{'type': 'potion', 'power': 10, 'price': 50}
{'type': 'potion', 'power': 10, 'price': 50}
...
{'type': 'sword', 'power': 50, 'price': 200}
{'type': 'sword', 'power': 50, 'price': 200}
{'type': 'potion', 'power': 10, 'price': 50}
{'type': 'potion', 'power': 10, 'price': 50}
{'type': 'potion', 'power': 10, 'price': 50}
...
```

List in dict

In [44]:
```
pizza={
    'crust':'thick',
    'toppings':['mushroom','vegetables','extra cheese']
}

print("You ordered a "+pizza['crust']+"-crust pizza "+"with the following toppin
for topping in pizza['toppings']:
    print("-"+topping)
```

```
You ordered a thick-crust pizza with the following toppings:
-mushroom
-vegetables
-extra cheese
```

In [45]:
```
# 薄弱题
favorite_languages={
    'jen':['python','ruby'],
    'sarah':['c'],
    'edward':['ruby','haskell']
}

for k,v in favorite_languages.items():
    print("\n"+k.title()+"'s favorite language are:")
    for v in favorite_languages[k]:
        print("-" + v.title())
```

```
Jen's favorite language are:
-Python
-Ruby

Sarah's favorite language are:
-C

Edward's favorite language are:
-Ruby
-Haskell
```

In [46]:

```python
# Excercise
hobbies={
    'alice':['reading','running'],
    'bob':['swiming'],
    'charie':['gaming','hiking','paiting']
}

for name in hobbies.keys():
    print("\n"+name.title())

    for hobby in hobbies[name]:
        print(" -"+hobby.title())
```

```
Alice
 -Reading
 -Running

Bob
 -Swiming

Charie
 -Gaming
 -Hiking
 -Paiting
```

In [48]:

```python
favorite_foods = {
    'mike': ['pizza', 'noodles'],
    'lily': ['sushi'],
    'tom': ['hamburger', 'cake', 'salad']
}

for name in favorite_foods.keys():
    print("\n"+name.title())

    for food in favorite_foods[name]:
        print("-"+food.title())
```

```
Mike
-Pizza
-Noodles

Lily
-Sushi

Tom
-Hamburger
-Cake
-Salad
```

dict in dict

In [49]:
```
users={
    'aeintein':{
        'first':'albert',
        'last':'einstein',
        'location':'princeton'
    },
    'mcurie':{
        'first':'marie',
        'last':'curie',
        'location':'paris'
    },
}

for username,user_info in users.items():
    print("\nUsername: "+username.title())

    full_name=user_info['first'] +" " + user_info['last']
    location=user_info['location']

    print("full_name: "+full_name.title())
    print("location: "+location.title())
```

```
Username: Aeintein
full_name: Albert Einstein
location: Princeton

Username: Mcurie
full_name: Marie Curie
location: Paris
```

In [99]:
```
# 薄弱题
# 6-7 dict in list
person1 = {
    'first_name': 'zhang',
    'last_name': 'san',
    'age': 20,
    'city': 'beijing'
}

person2 = {
    'first_name': 'li',
    'last_name': 'si',
    'age': 22,
    'city': 'shanghai'
}

person3 = {
    'first_name': 'wang',
    'last_name': 'wu',
    'age': 21,
    'city': 'guangzhou'
}

people=[person1,person2,person3]
```

```
for person in people:
    print(person)
```

```
{'first_name': 'zhang', 'last_name': 'san', 'age': 20, 'city': 'beijing'}
{'first_name': 'li', 'last_name': 'si', 'age': 22, 'city': 'shanghai'}
{'first_name': 'wang', 'last_name': 'wu', 'age': 21, 'city': 'guangzhou'}
```

In [116…

```
for person in people:
    print("\nInformation:")
    for k,v in person.items():
        print("  -"+k+":"+str(v))
```

```
Information:
  -first_name:zhang
  -last_name:san
  -age:20
  -city:beijing

Information:
  -first_name:li
  -last_name:si
  -age:22
  -city:shanghai

Information:
  -first_name:wang
  -last_name:wu
  -age:21
  -city:guangzhou
```

In [61]:

```
# 6-8 dict in list
pet1={'kind':'cat','owner':'wenda'}
pet2={'kind':'dog','owner':'lily'}
pet3={'kind':'bird','owner':'ana'}
pets=[pet1,pet2,pet3]

for num, pet in enumerate(pets, start=1):
    print(f"\npet{num} informations:")
    print("Kind of nanimal:"+pet['kind'].title())
    print("Name of owner:"+pet['owner'].title())
```

```
pet1 informations:
Kind of nanimal:Cat
Name of owner:Wenda

pet2 informations:
Kind of nanimal:Dog
Name of owner:Lily

pet3 informations:
Kind of nanimal:Bird
Name of owner:Ana
```

In [62]:

```
# 6-8 dict in list
pet1={'kind':'cat','owner':'wenda'}
pet2={'kind':'dog','owner':'lily'}
pet3={'kind':'bird','owner':'ana'}
pets=[pet1,pet2,pet3]

for pet in pets:
    print("\npet informations:")
```

```
    print("Kind of nanimal:"+pet['kind'].title())
    print("Name of owner:"+pet['owner'].title())
```

```
pet informations:
Kind of nanimal:Cat
Name of owner:Wenda

pet informations:
Kind of nanimal:Dog
Name of owner:Lily

pet informations:
Kind of nanimal:Bird
Name of owner:Ana
```

In [141…

```
# 6-9 list in dict
favorite_places={
    'mica':['chengdu','tianjin','lanzhou'],
    'sara':['hangzhou','sichuan','dalian'],
    'maria':['beijing','suzhou','hunan'],
}

for name in favorite_places.keys():
   print("\n"+name.title()+"'s favorite places are:")
   for place in favorite_places[name]:
       print(" -"+place.title())
```

```
Mica's favorite places are:
 -Chengdu
 -Tianjin
 -Lanzhou

Sara's favorite places are:
 -Hangzhou
 -Sichuan
 -Dalian

Maria's favorite places are:
 -Beijing
 -Suzhou
 -Hunan
```

In [152…

```
# 6-10 list in dict
favorate_num={
    "mica":[9,11,8],
    "donna":[5,8,3],
    "wang":[6,12,23],
    "sara":[7,9,4],
}

for name in favorate_num.keys():
    print("\n"+name.title()+"'s favorate number are:")
    for num in favorate_num[name]:
        print(" -"+str(num))
```

```
Mica's favorate number are:
 -9
 -11
 -8

Donna's favorate number are:
 -5
 -8
 -3

Wang's favorate number are:
 -6
 -12
 -23

Sara's favorate number are:
 -7
 -9
 -4
```

In [183…

```python
# 6-11 dict in dict
cities={
    'beijing':{'Country':'China','Population':'5 million','Fact':'capital of Chi
    'kuala lumpur':{'Country':'Malaysia','Population':'2 million','Fact':'a hot
    'newyork':{'Country':'America','Population':'3.5 million','Fact':'a develope
}

for city_name in cities.keys():
    print("\n"+city_name.title()+" has these info:")
    for key,value in cities[city_name].items():
        print(" -"+key+": "+value.title())
```

```
Beijing has these info:
 -Country: China
 -Population: 5 Million
 -Fact: Capital Of China

Kuala Lumpur has these info:
 -Country: Malaysia
 -Population: 2 Million
 -Fact: A Hot Weather City

Newyork has these info:
 -Country: America
 -Population: 3.5 Million
 -Fact: A Developed City
```

In [65]:

```python
# 6-12 dict in dict + 新增 key-value pair
cities={
    'beijing':{'Country':'China','Population':'5 million','Fact':'capital of Chi
    'kuala lumpur':{'Country':'Malaysia','Population':'2 million','Fact':'a hot
    'newyork':{'Country':'America','Population':'3.5 million','Fact':'a develope
}

cities['beijing']['food'] = 'Peking Duck'
cities['kuala lumpur']['food'] = 'Nasi Lemak'
cities['newyork']['food'] = 'Pizza'

for city_name, info in cities.items():
    print("\n"+city_name.title()+" :")
```

```
    for key, value in info.items():
        print("-" + key + ":"+ value)
```

```
Beijing :
-Country:China
-Population:5 million
-Fact:capital of China
-food:Peking Duck

Kuala Lumpur :
-Country:Malaysia
-Population:2 million
-Fact:a hot weather city
-food:Nasi Lemak

Newyork :
-Country:America
-Population:3.5 million
-Fact:a developed city
-food:Pizza
```

In [68]:

```
# 6-12
kuala_lumpur={'Country':'Malaysia','Population':'2 million','Fact':'a hot weathe
kuala_lumpur['food']='nasi goreng'

for k,v in kuala_lumpur.items():
    print(k.title()+": "+v)
```

```
Country: Malaysia
Population: 2 million
Fact: a hot weather city
Food: nasi goreng
```

In [ ]:

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